

1 The scientific method

EXPERIMENT 1

The scientific method

Goal

The goal of this experiment is to understand how the scientific method works. To do this you will make observations, look for patterns and come up with a hypothesis to *identify an unknown compound*. You will do this by comparing the reaction of the unknown compound with a set of known-reference-compounds, also called blanks. In this experiment, you will obtain some blank chemicals and an unknown. Your unknown is the same as one of the blanks. You will test each of these chemicals with a set of reagents. You have to observe the result of the test and record any change, such as *gas evolution*, the formation of a *precipitate*, a *change of color*, or maybe *no reaction*. Simultaneously, you will run similar tests for the unknown sample that is identical to one of the blanks. You will determine the unknown's identity by matching its characteristic reactions with one of the blanks.

Materials

- | | |
|--|---|
| ☐ Six test tubes. | ☐ Unknown compound. |
| ☐ The blanks: NaCl(s), NaI(s), Na ₂ CO ₃ (s), Na ₂ HPO ₄ (s) and Na ₂ SO ₄ (s). | ☐ The reagents: HNO ₃ (aq), Ba(NO ₃) ₂ (aq), AgNO ₃ (aq) and thymol blue. |

Experiment

Consider a set of Lego® blocks with different sizes and colors. We can combine them in numerous ways to construct many different structures. Similarly, if those blocks were atoms, we would obtain different molecules from the same set of atoms. Isomers are molecules with the same number and nature of atoms, and therefore the same molecular formula, but with different structures. Diverse structures arise from variations in the bonds between atoms and lead to new molecules, with different names and properties.

Geometric isomers are a subclass of isomers where not only the number and nature of the atoms are the same, but also the bonds. In this case, it is the spatial distribution or 3D structure that differs. This little change can lead to considerable differences in basic properties such as melting points and solubilities. In molecules with four ligands distributed around a central atom or a double bond, the isomerism can arise from the distribution of those four ligands around the molecule, leading to *cis*- and *trans*- isomers. This is the case in the proposed experiment.

The process of **isomerization** is the transformation from one molecule into its corresponding isomer. This transformation does not take place spontaneously but typically requires external energy and, in some cases, specific conditions. Energy is necessary to break certain bonds to allow the molecule to restructure into the corresponding isomer.

Procedure

Part A: Indicate your hypothesis

- ☐ *Step 1:* – Obtain the unknown compound, the blanks, and the reagents.

- ☐ **Step 2:** – Compare the unknown with the blanks and guess which blank looks like the unknown. That is your hypothesis. Write down the name of your guess on the results page.
- ☐ **Step 3:** – Wash a 100 mL graduated cylinder and 400 mL beaker. Place a pea-size portion of the unknown in the beaker, and add 150 mL of water using the cylinder to measure it. Swirl until all the solid has been totally or partially dissolved.
- ☐ **Step 4:** – Set aside the remaining portion of your unknown.

Good Lab Practice

 Obtain directions from your lab instructor for discharging the solutions that you will use in this experiment.



Part B: Testing for gas

- ☐ **Step 1:** – In this first experiment you will use the solid version of the blanks, not the droppers. Obtain six test tubes. In each test tube, you will add a blank reagent in a solid state, that is, one blank per test tube. To each test tube, you will add a pea-size sample of a different blank in a solid state using its spatula. You will add the unknown in the sixth test tube. This way, place a pea-size solid sample of NaCl in the first test tube. Add a pea-size sample of NaI in the second test tube, and so on with the rest of the blanks, Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 and your solid unknown in the following test tubes. Mark your test tubes for recognition.
- ☐ **Step 2:** – Add five drops of HNO_3  to each test tube and record your observations in the results table: *GE*, for gas evolution, *NGE* for no gas evolution.
- ☐ **Step 3:** – Discharge the solutions from the test tubes into the right waste container. Wash the test tubes, and rinse them with distilled water.

Part C: Testing for a barium precipitate

- ☐ **Step 1:** – For the following test will use only liquid samples in the dropper, not the powders. Use the solutions of NaCl, NaI, Na_2CO_3 , Na_2HPO_4 and Na_2SO_4 found in the laboratory, and the solution of your unknown prepared in Step 3 for the remaining tests in this practice.
- ☐ **Step 2:** – Using six clean test tubes, add 20 drops of each of the solutions as well as the unknown to separate test tube. In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution. Add 3 drops of a solution of ammonia (NH_3) in each test tube  and swirl until solved.
- ☐ **Step 3:** – Add five drops of the $\text{Ba}(\text{NO}_3)_2$  solution to each of the test tubes. Shake gently to obtain homogeneity. Examine each test tube carefully and look at the results. Record your observations: *P* for precipitate, *NP* for no precipitate.

CAUTION!

-  Nitric acid (HNO_3) and ammonia (NH_3)  can cause chemical burns.
-  Solutions containing barium are toxic.



Part D: Testing for silver precipitate

- ☐ **Step 1:** – Using six clean test tubes, add 20 drops of each of the solutions including the unknown to separate test tubes (e.g. NaCl, NaI, Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 , and unknown). In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution.

- ☐ *Step 2:* – Add five drops of the nitric acid AgNO_3 solution to each test tube. Record your observations: *P for precipitate, NP for no precipitate*. Include the color of the precipitate.

Part E: Testing with thymol blue

- ☐ *Step 1:* – Using six clean test tubes, add 20 drops of each of the solutions including the unknown to separate test tubes (e.g. NaCl , NaI , Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 , and unknown). In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution.
- ☐ *Step 2:* – Add five drops of the thymol blue solution to each test tube. Shake each test tube gently before recording your observations: *write down Blue, Yellow green, or Red*.

Part F: Identifying the unknown compound

- ☐ *Step 1:* – The unknown is one of these compounds: NaCl , NaI , Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 . You should be able to identify your unknown compound by matching its reactions with those of one of the known samples.
- ☐ *Step 2:* – When done with the experiment, clean up all test tubes using a test tube brush and soapy water.

STUDENT INFO

Name: _____

Date: _____

Pre-lab Questions**The scientific method**

1. Classify the following compounds as *ionic, covalent, acids, bases* or *oxo salts*:

NH ₃	_____	SO ₃	_____
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HNO ₃	_____	NaF	_____
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2. Give the names or formulas for the following compounds:

NaCl	_____	HNO ₃	_____
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Sodium iodide	_____	Sodium carbonate	_____
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3. Name or give the formula of the following compounds:

SO ₂	_____	NO ₂	_____
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Lithium fluoride	_____	Sodium oxide	_____
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4. What does a pea size of solid mean?

5. Where in the lab you will be disposing of the disposals from this experiment?

6. The following table resulted from an experiment, very similar to the one you are going to carry. The goal was to identify an unknown compound between KCl and KI. Look at the reactivity patterns and identify the unknown.

		Reagents			
		HNO ₃	Ba(NO ₃) ₂	AgNO ₃	Thymol Blue
Blanks	KCl	NP	NP	P _(white)	Green
	KI	NP	NP	NP	Green
	Unknown	NP	NP	NP	Green

My unknown is = _____

STUDENT INFO

Name: _____

Date: _____

**Results
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Unknown Label # = _____ Your guess = _____

		Reagents			
		HNO ₃	Ba(NO ₃) ₂	AgNO ₃	Thymol Blue
Blanks	NaCl				
	NaI				
	Na ₂ CO ₃				
	Na ₂ HPO ₄				
	Na ₂ SO ₄				
	Unknown				

(write GE for gas evolution, NGE for no gas evolution, P for precipitate, NP for no precipitate and Blue, Yellow or Green for the color for the Thymol blue experiment)

My unknown is = _____

STUDENT INFO

Name: _____

Date: _____

Post-lab Questions

The scientific method

1. Classify the following compounds as *ionic*, *covalent*, *acids*, *bases* or *oxo salts*:

Ba(OH)₂

CO₂

HCl

H₂SO₄

LiCl

AuCl

Na₂SO₄

NaOH

2. Give the names or formulas for the following compounds:

HNO₃

Sodium sulfate

Barium nitrate

Na₂CO₃

AgNO₃

Nitric acid

3. Name or give the formula of the following compounds:

CuCl₂

Hydrogen chloride

Nickel (II) chloride

BaO

CoO

MgO

Sodium iodide

Potassium iodide

NH₃

LiH
